

Probability in Closed Symbolic Systems: When Classical Probability Proves What Statistics Cannot

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Abstract

This paper develops a framework for using classical probability, rather than frequentist statistics, to assess relatedness and origins in closed symbolic systems: domains characterized by a small finite alphabet, well-defined combinatorial rules, and effective closure against external input. Frequentist statistics is designed for inference from repeated sampling of an open or incompletely specified population; classical probability is designed for reasoning about a single case within a fully specified, finite universe of possibilities. When a domain meets the criteria for closure, classical probability provides a more direct, transparent, and philosophically appropriate tool than statistical significance testing — particularly for singular, unrepeatable events for which no sample can be drawn. The framework is illustrated with two extended case studies: the identity between the Proto-Indo-European copula root and third-person pronoun root, and the extreme conservation of ultraconserved non-coding elements in vertebrate genomes. Further applications — to Hebrew computational linguistics, language acquisition, and the evaluation of singular communicative events such as a pet parrot's unprompted, contextually accurate alarm call — extend the framework's reach across historical linguistics, comparative genomics, and cognitive science.

1. Introduction

In the early morning hours of February 3, 2020, in Lebanon, Tennessee, Barbara Klein, her husband Larry, and their six-year-old granddaughter were asleep when a fire broke out in their kitchen. They were awakened by their pet parrot, Louie, frantically and repeatedly squawking the word “Fire! Fire!” — a word the family reported he had never uttered before. Thanks to Louie’s urgent alarm, the family escaped with their lives. Tragically, Louie and several family dogs perished in the blaze (SWNS 2020).

This was a singular, unrepeatable event: a precise, contextually appropriate vocalization at exactly the right moment. Crucially, the Klein family reported that Louie had never uttered the word “Fire!” before that night, under any context. Had “fire” simply been an existing word in his repertoire, the relevant question would be the probability of selecting that one word from his known vocabulary. But since it was not, a different and more demanding question applies: what is the probability of a parrot spontaneously producing a novel phoneme sequence that happens to constitute a meaningful, contextually correct English word — without that sequence having been previously learned or rehearsed?

This is the appropriate question, because the vocalization must be evaluated against the phoneme inventory of the target language, English, not against any inherent limit on what sounds a parrot can produce. Parrots are capable of an extremely wide range of vocalizations; the relevant constraint is that “fire” itself, to count as that specific word, must be assembled from the roughly 44 phonemes of English. The word “fire” in American English consists of three phonemes: /f/, /aɪ/ (the diphthong), and /ə/ (the r-colored schwa). Under uniform distribution across 44 phonemes, the probability of producing this specific three-phoneme sequence by chance is:

$$P(\text{“fire”}) = 1/44 \times 1/44 \times 1/44 \approx 1/85,000$$

Phonotactic refinements — accounting for the relative frequency of /f/ in initial position, /aɪ/ among vowels and diphthongs, and /ə/ in coda position — push in different directions and roughly cancel, leaving approximately 1/85,000 as a defensible single estimate. This is the probability of producing the phoneme sequence constituting “fire” by chance, without any training, in a context where it happens to be precisely the right word. No null hypothesis is required, no repeated trials, no experimental design. The universe of possible phoneme sequences is finite and enumerable; the probability is directly calculable from the structure of the system.

The contrast with a more famous parrot illuminates what is at stake. Alex, the African Grey parrot studied for decades by researcher Irene Pepperberg, was trained through hours of daily interaction to answer questions about colors, shapes, and quantities. Pepperberg used statistical methods to demonstrate that Alex’s correct responses

exceeded chance rates (Pepperberg 1999) — an important and genuine contribution to comparative cognition. But precisely because Alex was trained, rewarded, and repeatedly tested, the statistical demonstration leaves open a question that the training cannot resolve: how much of Alex’s performance reflected genuine understanding and how much reflected sophisticated operant conditioning? Repeated sampling in a controlled environment is the right tool for ruling out chance in a trained animal. It cannot distinguish understanding from conditioning, because both produce the same observable pattern under those conditions.

Louie’s case is structurally different. There was no training, no reward, no experimental protocol. The word “Fire!” had never been uttered by Louie before. The probability that he produced it spontaneously, twice, at precisely the moment it was needed, by chance, is approximately $1/85,000$ — a figure derivable directly from the phoneme inventory of his vocal system, without any statistical machinery. **This is significant — not statistically significant — but actually significant.** The spontaneity is not a methodological weakness; it is the evidential point. Unrepeatable, unprompted, unrewarded productions are precisely the events that most clearly separate understanding from conditioning — and classical probability is the appropriate tool for evaluating them, because they are single cases within a closed system rather than samples from an open population.

Scientists trained primarily in frequentist statistics often object that inferences about singular unrepeatable events cannot constitute rigorous evidence. “You have to set up an experiment that produces replicable results,” the argument goes, “otherwise there is no proof.” This stance reflects the heavy emphasis on statistical inference in university curricula. Probability is typically taught, if at all, only as a precursor to statistics, with its classical roots in reasoning about single cases largely overshadowed. Yet classical probability, as developed by Pascal, Fermat, and Laplace, was designed precisely for assessing specific outcomes within a fully specified, finite universe of possibilities — without any requirement for repeated trials.

This paper develops a framework for assessing relatedness and origins in closed symbolic systems: domains characterized by a small finite alphabet, well-defined combinatorial rules, and effective closure against external input at the relevant level of analysis (Section 2). In such systems, direct probability calculations often provide a more transparent and appropriate measure of surprise than frequentist statistical methods, particularly for unrepeatable historical events. The framework is illustrated with two extended case studies: one from historical linguistics involving pronoun-copula root identity in Proto-Indo-European (Section 4; see also Katz, LingBuzz preprint), and one from comparative genomics involving ultraconserved non-coding elements in vertebrate genomes (Section 5; Bejerano et al. 2004). Broader implications — for natural language processing, language acquisition, and the evaluation of singular communicative events — are explored in Section 6.

The central methodological claim is that distinguishing probability — reasoning about a single case within a known closed system — from statistics — inference from repeated sampling of an open or incompletely characterized population — clarifies inferential practice across disciplines (Hacking 2001). The distinction is not merely terminological. It determines which questions can be asked, which answers are available, and which inferential errors are most likely to occur when the wrong framework is applied to a problem that the other framework is designed to handle.

2. Two Kinds of Inferential Questions

The distinction between probability and statistics is not merely terminological. The two frameworks address fundamentally different kinds of questions, and deploying the wrong one produces answers that are technically valid but conceptually mismatched to the problem at hand.

2.1 What Statistics Does

Statistical inference is designed for situations where we have observed a sample drawn from a larger, partially unknown population, and we wish to draw conclusions about that population from what the sample reveals. The characteristic question of statistics is: given these observations, what can we infer about the process that generated them?

The machinery of frequentist statistics — null hypotheses, p-values, confidence intervals, significance thresholds — is built around this sampling relationship. A p-value, for instance, answers the question: if the null hypothesis were true, how often would we observe results at least as extreme as these, across repeated hypothetical samples? The

answer is always conditional on a model of how data could have been generated differently, and it is always expressed in terms of a long-run frequency across many hypothetical repetitions of the experiment.

This framework is enormously powerful in its intended domain. Clinical trials, population genetics, survey research, experimental psychology — all involve drawing samples from populations that are too large or too variable to be fully specified in advance. Statistics is the right tool for these problems because the uncertainty is genuinely about population parameters that the sample only partially reveals.

2.2 What Probability Does

Classical probability, as developed by Pascal, Fermat, and Laplace, addresses a different kind of question: given a fully specified, finite universe of possible outcomes, what fraction of those outcomes constitute the event of interest? The characteristic question of classical probability is not about samples or populations at all — it is about the structure of a known, closed system.

The canonical examples are games of chance: dice, cards, coin flips. The universe of possible outcomes is fully enumerable. No sampling is required, no population need be inferred, no null hypothesis is needed. The probability of an event is simply the ratio of favorable outcomes to total outcomes, derivable by direct calculation from the known structure of the system.

Hacking (2001) distinguishes this classical framework from the frequentist interpretation of probability that underlies modern statistics, and traces the historical divergence between the two.¹ The distinction matters here because it identifies the conditions under which each framework is appropriate: statistics when the universe is incompletely known and must be inferred from samples; probability when the universe is finite, fully specified, and closed.

2.3 The Misapplication of Statistics to Closed Systems

The two frameworks are not interchangeable, and applying statistics to a problem that calls for probability produces a characteristic distortion. Statistical significance tells us how surprising an observation is against a background model — but when the background model itself is fully specifiable, we do not need the statistical machinery to answer the question. We can calculate the probability directly.

Consider the question of whether two three-letter sequences drawn from a thirty-symbol alphabet are identical by chance. The statistician might construct a null model of random sequence generation, observe a match, and report a p-value. The probabilist simply asks: what fraction of all possible three-symbol sequences match a given target? The answer —

$$1/30 \times 1/30 \times 1/30 = 1/27,000$$

— requires no sample, no null hypothesis, and no significance threshold. It is a direct fact about the structure of the system.

The statistical approach is not wrong in any formal sense — applied correctly, it will converge on the same answer. But it introduces unnecessary complexity, requires assumptions about the sampling process that may not apply, and produces an output (a p-value) that is frequently misinterpreted even by specialists. When the universe is closed and fully specifiable, direct probability calculation is simpler, more transparent, and more appropriate.

¹An information-theoretic framing offers a complementary, though not equivalent, perspective on the same underlying improbability; developing it fully is left for future work rather than substituted for the classical probability calculations used in the body of this paper. Shannon surprisal, defined as $-\log_2(p)$ for an event of probability p , gives a phoneme-level surprisal of approximately 16.4 bits for Louie's utterance, corresponding to the 1/85,000 figure discussed in Section 6.1. A word-level alternative can be constructed from the Oxford English Dictionary's widely cited count of 171,476 words in current use in its second edition (1989) — the standing published figure, since the third edition remains incomplete — giving a surprisal of approximately 17.4 bits under a uniform-selection model over the lexicon. These two figures are not additive: they answer different questions (phoneme-level production versus word-level selection from the lexicon) and are reported here only as complementary confirmations of the same underlying improbability, not as components of a single combined estimate.

2.4 The Criterion: When is a System Closed?

The question of which framework to apply therefore reduces to a prior question: is the relevant universe finite, fully specified, and closed? We propose the following criterion: a system is closed for purposes of probabilistic analysis when three conditions are met.

First, the system operates over a small, finite alphabet of discrete symbols — a set of possible elements that is enumerable in advance and does not grow during the analysis. Natural language phoneme inventories (typically 20–40 segments), DNA nucleotides (4), and the tokens of a formal writing system are examples. Open-ended vocabularies — the words of a living language, for instance — do not meet this criterion when considered diachronically, since new words enter and old words fall out of use continuously. A synchronic snapshot of such a vocabulary — for example, the word count recorded in a specific, dated dictionary edition — can be treated as a fixed, enumerable set for a bounded illustrative calculation (see the surprisal estimate in Section 2.2's footnote), but this is a distinct and weaker sense of closure than the diachronic closure required for the classical-probability arguments developed in Sections 4 and 5. Closure claims in this paper are therefore implicitly claims about a system at a specified point in time, or over a specified stable window, not about its behavior across its entire history — a distinction that also matters for a second reason: the 'external input' condition below concerns contact-driven input from outside a system (borrowing, horizontal gene transfer, copying from outside sources), not internal change (sound shifts, mergers, splits, reanalysis), which can alter a system's symbol inventory or combinatorial structure over time without any external contact at all.

Second, the system's combinatorial constraints are well-defined — the rules governing which symbol sequences are permissible are knowable and can be specified independently of the particular sequences under analysis. Phonotactic constraints in historical linguistics and base-pairing rules in molecular biology are examples.

Third, the system is reasonably closed against external input — new symbols cannot enter the system freely from outside during the period under analysis. This is the borrowing-resistance condition: the system must have a history during which its symbol inventory was stable enough that identity between two sequences can be assessed against a fixed background. Completely isolated systems are rare; the question is whether the degree of openness is sufficient to invalidate the probability calculation, or whether the categories under comparison — core vocabulary, non-coding conserved sequences — are sufficiently insulated from external input to treat the system as effectively closed.

When all three conditions are met, classical probability provides a direct, transparent, and interpretable measure of the likelihood of coincidental identity — and statistical significance testing, while not formally wrong, is a more complex and less appropriate instrument for the job.

3. Closed Symbolic Systems: A Definition

Section 2 proposed three conditions for treating a system as closed for purposes of probabilistic analysis: a small finite alphabet, well-defined combinatorial constraints, and reasonable closure against external input. This section develops those conditions more carefully, introduces the concept of degrees of closure, and draws out a consequence that has been largely overlooked in both linguistics and biology: the appropriate prior probability for any identity claim — whether to treat it as coincidence or as common origin — is not universal but is itself a function of how closed and systematic the relevant system is.

3.1 Closure as a Spectrum

No natural symbolic system is perfectly closed. Languages borrow; genomes experience horizontal gene transfer; texts are copied with errors and interpolations. The question is not whether a system is absolutely closed but whether it is closed enough — whether the degree of external influence on the relevant categories is small enough that a closed-system probability model gives a reasonable approximation of the actual likelihood of coincidental identity.

This means closure is not binary but exists on a spectrum. At one end are systems that are almost completely self-contained for the categories of interest: the consonantal root inventory of Classical Arabic or Biblical Hebrew, where borrowing into the closed-class root system is vanishingly rare and the combinatorial constraints are so tight that nearly every consonantal skeleton that appears in the lexicon belongs to a recognizable derivational family. At the other end are systems like the English lexicon, where borrowing from Latin, French, Greek, and Germanic

sources has produced a mixed inventory in which similar-sounding forms frequently have unrelated origins and the mapping between phonological form and etymological source is opaque to most speakers.

Between these extremes lie systems of intermediate closure: Mandarin, where the morphological system is highly transparent and systematic within the native stratum but phonological erosion has produced some homophones; Modern Greek, where ancient and modern strata coexist; and Proto-Indo-European itself, which is reconstructed on the assumption of a relatively systematic root-based grammar operating over a finite consonantal inventory.

3.2 Systematicity and the Prior

The position of a system on the closure spectrum has a direct consequence for probabilistic reasoning that has rarely been made explicit: it determines the appropriate prior probability for any identity claim.

In a highly systematic, closed root system, the prior probability that two sequences sharing the same symbolic content reflect the same underlying root is high. The system actively constrains the space of possible roots and makes the mapping between form and meaning transparent and productive. When two words in Biblical Hebrew share a consonantal skeleton, the default assumption — the prior with which a Semiticist approaches the data — is that they are etymologically related, and divergence from common origin requires explanation. This prior is not a cultural bias; it is a rational response to the structure of the system.

In a non-systematic or hybrid system like English, the prior runs in the opposite direction. Similar-sounding words are frequently unrelated — bare and bear, flour and flower, plum (from Proto-Germanic) and plumb (from Latin plumbum, lead) — because the system was assembled from multiple etymological strata with different phonological histories. The default assumption for an English-trained linguist encountering similar-looking forms is coincidence, and connection requires proof. This prior is equally rational given the structure of the system.

Neither prior is wrong in its home domain. The problem arises when scholars carry the prior from their home system into a domain where it does not apply. A Semiticist trained on a high-prior system will tend to see common origins where an English-trained linguist trained on a low-prior system will see coincidence — and vice versa. Without making the prior explicit and grounding it in the structure of the specific system under analysis, this clash of assumptions is invisible and unresolvable. Making the prior explicit is precisely what the probabilistic framework proposed in this paper enables.

3.3 An Empirical Illustration: The Spelling Bee

The typological conditioning of priors has observable consequences even within a single non-systematic language. Children who excel at competitive spelling in English do not succeed primarily by memorizing more words. They succeed by developing probabilistic intuitions about the etymological stratum to which an unfamiliar word belongs: whether its structure suggests a Latin origin (and therefore certain suffix patterns), a Greek origin (and therefore certain consonant clusters and combining forms), or a Germanic origin (and therefore different phonological regularities). Each etymological stratum functions, for this purpose, as a relatively closed subsystem with its own combinatorial constraints, and the skilled speller has learned to identify which subsystem applies and to apply the relevant prior.

This is classical probability reasoning applied to a partially closed system. The speller is not running a statistical test against a null hypothesis of random letter combination. They are reasoning from the known structure of a finite system — the set of Latin suffixes, the Greek combining forms, the Germanic sound correspondences — to the most probable form of an unfamiliar word. The same reasoning that allows a Semiticist to navigate unpointed Hebrew text, or that allows a historical linguist to reconstruct a proto-form from attested daughter language reflexes, is operating here: structured probabilistic inference from a known system, not statistical inference from a sample.

3.4 Implications for the PIE Case

The framework developed in this section bears directly on the PIE paper's argument. PIE, on the standard reconstruction, was a highly systematic root-based language with a small consonantal inventory, productive ablaut alternations, and transparent morphological derivation. By the typological criterion developed in Section 3.2, PIE should be assigned a relatively high prior for root identity — meaning that the uniform distribution assumption used in the probability calculation (which implicitly assigns a prior of 1/27,000 to any three-phoneme match) is already

conservative. A Semiticist approaching PIE data would be right, by the typological prior criterion, to find the copula-pronoun identity compelling even before the calculation is run; the calculation then makes explicit what the typological prior already suggests.

Conversely, the resistance to this argument from scholars trained primarily on European languages may partly reflect a low prior imported from a less systematic linguistic environment — a prior that is appropriate for English but not for PIE. Making the prior explicit does not resolve all disagreements, but it at least makes visible where they originate.

4. Case Study 1: Pronoun-Copula Root Identity in Proto-Indo-European

The first case study applies the closed symbolic systems framework to a specific question in historical linguistics: whether the apparent identity between the Proto-Indo-European copula root $*h_1és-$ and the third-person pronoun root reflects a genuine historical connection or a coincidence of reconstruction. The full argument, including daughter language paradigm tables, literature engagement, and sensitivity analysis, is developed in a companion paper (Katz, LingBuzz preprint). This section summarizes the key moves to illustrate how the framework of Sections 2 and 3 applies to a linguistic case.

4.1 PIE as a Closed Symbolic System

Proto-Indo-European meets all three conditions for closed-system probabilistic analysis established in Section 2.4. Its phonemic inventory — approximately 30 segments under the standard laryngealist reconstructions — is finite and well-documented across four major competing reconstructions (Beekes 1995; Fortson 2004; Meier-Brügger et al. 2010; Szemerényi 1980). Its combinatorial constraints are well-defined: PIE root phonology is governed by systematic phonotactic restrictions on consonant co-occurrence (Iverson and Salmons 1992; Corbeau 2013). And the categories under comparison — the copula and the third-person pronoun — are among the vocabulary items most resistant to borrowing cross-linguistically (Haspelmath and Tadmor 2009), satisfying the closure-against-external-input condition.

PIE also occupies the high-systematicity end of the closure spectrum described in Section 3.1. As a root-based language with productive ablaut alternations and transparent morphological derivation, it assigns a high prior to root identity — meaning that the default assumption for any two phonemically identical sequences is common origin rather than coincidence. The probability calculation does not create this expectation; it makes it explicit and quantifies it.

4.2 The Calculation

The copula root $*h_1és-$ and the third-person pronoun root are identical across three phonemes — $*h_1$, $*e$, and $*s$ — in the same order. Under uniform distribution across the 30-phoneme inventory, the probability of coincidental three-phoneme identity is:

$$P(\text{coincidental identity}) = 1/30 \times 1/30 \times 1/30 = 1/27,000$$

This baseline figure treats all phonemes as equally probable in all positions. A more realistic calculation accounts for the position-specific structure of PIE roots: the vowel position is nearly invariant (since $*e$ is the default ablaut vowel of the vast majority of PIE roots), and phoneme frequencies in initial and final consonant positions are not uniform. Using observed frequencies — $*h_1$ in approximately 3.2% of root-initial positions (Hartmann 2021), $*e$ in approximately 60% of root-medial positions as the dominant ablaut vowel, and $*s$ in approximately 12% of root-final positions — yields a frequency-adjusted estimate of:

$$P(\text{frequency-adjusted}) = 0.032 \times 0.60 \times 0.12 \approx 1/13,000$$

Both the baseline and frequency-adjusted figures are well below any credible threshold for coincidence. The identity is further confirmed within the PIE oblique pronoun paradigm itself — the genitive $*h_1éso$, dative $*h_1esmōi$, ablative $*h_1esmōd$, and locative $*h_1esmi$ all show the full three-phoneme root — and persists independently across

six daughter branches: Sanskrit, Greek, Avestan, Latin, Hittite, and Old Irish. Hittite, the most archaic attested branch, provides direct attestation of the laryngeal rather than requiring its reconstruction from secondary effects.

4.3 What This Case Illustrates

The PIE case illustrates several features of the closed-system framework that generalize beyond this specific example.

First, the distinction between probability and statistics is not merely theoretical here — it is practically decisive. There is no sample to draw from PIE, no population to generalize to, no repeated trials available. The standard comparative method, which requires written attestation of an earlier stage, cannot resolve questions internal to the reconstructed system itself. Classical probability can, because it requires only a finite, well-specified universe — which PIE's phonemic inventory provides.

Second, the typological prior matters. As Section 3.2 established, PIE's high systematicity means that root identity should be treated as the default rather than as something requiring proof. The probability calculation confirms this prior quantitatively, but the prior itself is grounded in the typological character of the language — a point that explains why scholars trained on more systematic languages (Semiticists, in particular) have often found the copula-pronoun connection intuitively obvious, while scholars trained on less systematic languages have required the explicit calculation to find it compelling.

Third, the case illustrates that the framework is robust across methodological choices. A sensitivity analysis across four competing PIE reconstructions and a range of phoneme frequency assumptions shows that the conclusion — coincidental identity is not a credible explanation — holds under all plausible parameter values (Katz, LingBuzz preprint, Section 4.3).

5. Ultraconserved Non-Coding Elements in Vertebrate Genomes

Among the most striking findings in comparative genomics is the existence of sequences that are nearly identical across species separated by hundreds of millions of years of independent evolution. These ultraconserved elements (UCEs) and conserved non-coding elements (CNEs) — stretches of DNA often 50 to 200 or more base pairs long — remain essentially unchanged from fish to human, despite the vast evolutionary distances involved (Bejerano et al. 2004; Pennacchio et al. 2006). The standard biological response is to note their statistical significance against a background model of neutral mutation. The framework developed in this paper suggests a more direct approach.

The genetic code exemplifies a closed symbolic system in the strictest sense. DNA employs a four-letter alphabet {A, C, G, T} — adenine, cytosine, guanine, and thymine — whose specific composition is not arbitrary but arises from fundamental biochemical constraints. The four bases form two complementary pairs (A–T and G–C) through hydrogen bonding, enabling stable replication and the double-helix structure (Watson and Crick 1953). From an information-theoretic perspective, four letters provide an efficient balance: 2 bits of information per position and, when read in triplets, 64 possible codons — sufficient to encode 20 amino acids with built-in redundancy for error correction (Freeland and Hurst 1998). Synthetic biology experiments have shown that expanding or reducing this alphabet tends to compromise either chemical stability or coding capacity (Szathmáry 2003). The four-letter DNA code is, in the terms of Section 2, a small finite alphabet with well-defined combinatorial constraints and maximal closure against external input — no horizontal gene transfer can introduce a fifth nucleotide. It is the most closed symbolic system considered in this paper.

The probability calculation follows directly from the structure of the system. For a conserved sequence of length n , the classical probability of an exact match arising independently in two lineages is $(1/4)^n$ — a direct calculation from the four-letter alphabet, requiring no sampling assumptions. Applied to the sequence lengths reported by Bejerano et al. (2004), this yields probabilities on the order of 10^{-12} for a modest 20-base-pair element, and 10^{-22} or lower for the longer UCEs documented in that study. Refinements using observed base frequencies, dinucleotide constraints, and the known regulatory syntax of non-coding DNA only lower these figures further. As Section 3 established, the genetic system occupies the closed end of the closure spectrum — meaning the prior probability of common origin for any identical sequence is already very high before the calculation is run. The calculation then makes explicit

what the prior already strongly suggests: vertical transmission from a common ancestor, maintained by purifying selection, is not merely the most parsimonious explanation. It is the only explanation consistent with the probability of the data.

The parallel with the linguistic argument is precise. In both domains we observe a small finite alphabet, strong combinatorial and syntactic constraints on permissible sequences, and highly conserved core elements whose probability of coincidental identity is too low to be credible under independent origin. The standard statistical treatment in both fields — reporting significance against a background model — answers a different question than the one that matters: not “is this result surprising given a null model?” but “what is the probability that this specific sequence arose independently?” Classical probability answers the second question directly, without the sampling assumptions that statistical significance testing requires. Public multi-species genome alignments — UCSC Genome Browser, Ensembl Compara, phastCons tracks — make such analyses fully reproducible, extending the approach naturally from historical linguistics to evolutionary biology without requiring new sequencing data.

5.1 Why the Difference in Method Matters

Both approaches — the statistical methods of Bejerano et al. (2004) and the classical probability framework proposed here — reach the same broad conclusion: UCEs are extraordinarily unlikely to have arisen independently in separate lineages. The difference lies not in the direction of the conclusion but in its epistemic grounding, transparency, generality, and philosophical robustness.

Statistical models estimate improbability based on observed evolutionary processes across lineages — neutral substitution rates, phylogenetic comparisons, branch-length calibrations. They are powerful tools, but they remain tied to empirical rate estimates that can be noisy, model-dependent, or contested. Different neutral rate calibrations, variation in mutation spectra, and uncertainty about ancestral population sizes all introduce parameters whose values must be assumed rather than derived. The conclusion is therefore conditional on the model: UCEs are surprising under this particular model of neutral evolution.

Closed-system probability starts from the intrinsic structure of the system itself: a four-letter alphabet plus known combinatorial and functional constraints on regulatory sequences. It directly calculates the size of the possibility space without assuming a particular evolutionary model, branch lengths, or pattern of repeated sampling. This gives a more fundamental answer to why long identical sequences are improbable — not merely “they rarely occur in alignments under neutrality” but “there are astronomically many possible 200-base-pair sequences in this four-letter alphabet, and the probability that two independently evolving lineages would arrive at the identical sequence by chance alone is essentially zero.” A sensitivity analysis, of the kind performed for the PIE **h₁és-* case in the companion paper, can explore the full range of assumptions transparently and without hiding them inside a model.

The classical-probability calculation addresses coincidental origin under a model with no constraint at all — but a more sophisticated objection asks whether strong purifying selection could independently drive two unrelated lineages to converge on an identical functional sequence, without common ancestry. This objection does not apply to UCEs as identified: they are defined as orthologous genomic loci — the same chromosomal position, flanked by the same syntenic context — across species whose relatedness is already independently established from thousands of unrelated genomic regions. Common ancestry of the locus itself is therefore not in question; what remains to be explained is why its content has stayed identical for 300–400 million years, and it is this question the probability calculation and the purifying-selection hypothesis both address. This maintenance mechanism is itself still debated: reduced mutation rate (‘mutational cold spot’) and strong ongoing purifying selection remain competing explanations, and estimates of the strength of that selection have varied considerably across studies (Katzman et al. 2007). Direct functional knockout studies, moreover, have found several UCEs individually dispensable for organismal viability, deepening rather than resolving the puzzle of their extreme conservation.

The closed-system framework also offers greater generality across disciplines. As this paper demonstrates, the same inferential logic applies in historical linguistics, comparative genomics, and natural language processing. Statistical models are powerful but domain-specific in their application; classical probability in closed systems reveals a common structure wherever finite alphabets and strong combinatorial rules govern the generation of sequences. This cross-disciplinary portability is a genuine methodological contribution, independent of the specific results it produces in any one domain.

Finally, as Hacking (2001) argues, the classical probability framework restores a philosophically important distinction: reasoning about a single, fully specified case within a known universe is not the same as frequentist inference about populations. Many scientific debates become clearer once it is recognized that the system under study is a closed symbolic system rather than a sample from an open one. Adopting this perspective opens new questions that the statistical approach tends to foreclose: What are the effective combinatorial constraints on enhancer grammar? How small does the functional possibility space need to be for conservation across 400 million years to be expected under common ancestry? How do these constraints interact with the prior probability of identity established by the typology of the system? These questions are tractable within the closed-system framework in a way they are not within standard comparative statistical genomics.

The statistical result tells us that UCEs are surprising under neutrality. The closed-system probability framework tells us why they are surprising in a way that generalizes across biology, language, and symbolic systems — and does so with greater conceptual economy.

6. Implications and Extensions

The closed symbolic systems framework has broad implications beyond the two main case studies. It reframes how we evaluate evidence in domains where frequentist statistics is least applicable: singular, unrepeatable historical events within finite, structured possibility spaces.

6.1 One-Time Events and the Limits of Replicability

The Louie the parrot case (February 2020) provides a vivid illustration. A pet parrot spontaneously produced the novel utterance “Fire! Fire!” — a word its owners insist it had never said before — precisely when a real fire threatened a sleeping family. From a phoneme-level perspective, the probability of this specific sequence emerging by chance is on the order of 1/85,000 (or lower when accounting for phonotactic constraints, repetition, and contextual urgency). Lay observers tend to feel this improbability intuitively, while many scientists steeped in statistical training may miss or downplay that intuitive sense.

The contrast with Alex the African Grey parrot, studied for decades by Irene Pepperberg, is instructive. Alex was trained through hours of daily interaction to answer questions about colors, shapes, and quantities, and Pepperberg used statistical methods to demonstrate that his correct responses exceeded chance rates. But precisely because Alex was trained, rewarded, and repeatedly tested, the statistical demonstration cannot resolve the question it most needs to answer: how much of Alex’s performance reflected genuine understanding and how much reflected sophisticated operant conditioning? Repeated sampling in a controlled environment is the right tool for ruling out chance in a trained animal. It cannot distinguish understanding from conditioning, because both produce the same observable pattern under those conditions.

Louie’s case is structurally different. There was no training, no reward, no experimental protocol. The spontaneity is not a methodological weakness; it is the evidential point. Unrepeatable, unprompted, unrewarded productions are precisely the events that most clearly separate understanding from conditioning — and classical probability is the appropriate tool for evaluating them, because they are single cases within a closed system rather than samples from an open population.

6.2 Hebrew Writing and the Costs of Statistical Thinking: A Cautionary Case

A particularly instructive illustration of what goes wrong when statistical reasoning is applied to a closed symbolic system comes from computational linguistics — specifically, from two decades of difficulty in processing Modern Hebrew text automatically.

Written Hebrew presents an unusual feature from the perspective of natural language processing: the standard orthography omits vowels entirely. Fluent readers encountering an unvoweled Hebrew text do not experience this as missing information. The consonantal skeleton of each word, combined with its morphological pattern, syntactic position, and semantic context, constrains the possible readings so tightly that vowels are recoverable with high accuracy. This is possible precisely because Hebrew morphology is a closed symbolic system: a small inventory of

consonantal roots — typically three consonants — combined with a finite set of derivational and inflectional templates generates the vocabulary, and the mapping between consonantal skeleton and meaning is highly systematic. Fluent readers are not guessing; they are calculating, implicitly, from a fully specified combinatorial system.

For decades, computer scientists attempting to build Hebrew NLP systems treated this as a statistical disambiguation problem: given an unvoweled word form, train a model on observed frequencies and predict the most likely reading. Wintner (2003) surveyed the state of Hebrew computational linguistics and identified the core difficulty: the writing system, rich morphology, and unique root-and-pattern word formation process made standard NLP pipelines designed for English — which treat word forms as the basic unit — fundamentally inadequate for Hebrew. The problem was not insufficient data; it was a category error in the inferential framework.

Tsarfaty, Sadde, Klein, and Seker (2019) diagnosed the same problem from a machine learning perspective a decade and a half later: standard annotation pipelines failed on Hebrew because errors in early morphological disambiguation decisions propagated through the pipeline and could not be recovered later. Their solution — joint inference over morphology and syntax rather than sequential statistical processing — is, from the perspective of this paper, a recognition that Hebrew requires reasoning about a closed system simultaneously rather than sampling from surface frequencies sequentially. The breakthrough was not more data or better statistics; it was a different model of what kind of problem Hebrew parsing actually is.

6.3 Language Acquisition and the Poverty of the Stimulus

Chomsky’s poverty of the stimulus argument holds that children acquire complex grammar from input that is too limited and degenerate to support induction by purely domain-general mechanisms. This has motivated positing a rich, domain-specific Language Acquisition Device embodying Universal Grammar.

An alternative — or complementary — view is that children leverage built-in probabilistic reasoning optimized for closed symbolic systems. The brain treats language as having a small finite phoneme alphabet, strong combinatorial and hierarchical constraints, and effective closure at the grammatical level. Within this bounded space, even sparse positive evidence supports rapid, accurate inference. Bayesian models of acquisition demonstrate that rational probabilistic updating with modest structural priors can resolve many classic poverty-of-stimulus problems without requiring highly language-specific innate rules (Xu and Tenenbaum 2007; Perfors et al. 2006). The Language Acquisition Device may thus be less a repository of fixed syntactic principles and more a domain-general probabilistic engine specialized for finite, rule-governed symbolic domains.

6.4 Methodological and Cross-Disciplinary Implications

The distinction between probability and statistics clarifies when each tool is appropriate. Fields dealing with singular historical events — historical linguistics, comparative genomics, and the evaluation of spontaneous communicative behavior — benefit from explicit closed-system probability calculations. Where the universe of possibilities is finite, structured, and known in advance, classical probability provides a cleaner, more transparent, and more principled answer than statistical significance testing. Recognizing this boundary is not a critique of statistics but a clarification of its scope.

This convergence is visible even in domains built on statistical machinery. Large language models are trained by statistical means — gradient descent over massive corpora, optimizing a probability distribution across billions of parameters. But inference at generation time is a different operation: at each step, the model selects the next token from a fixed, finite, fully enumerable vocabulary via a softmax distribution over that closed set. The training process is statistical; the moment of token selection, given the trained distribution, is a classical probability event over a closed symbolic system in exactly the sense developed in this paper. The two frameworks are not in competition here — they operate at different stages of the same system, and distinguishing them clarifies what each stage can and cannot explain.

7. Conclusion

This paper has argued that many important questions in the historical and cognitive sciences involve closed symbolic systems — domains with finite alphabets, well-defined combinatorial rules, and effective closure at the

relevant level of analysis. In such systems, classical probability provides a more natural and transparent tool for assessing surprise and relatedness than frequentist statistical methods, particularly for singular, unrepeatable events.

The two main case studies illustrate the point. In Proto-Indo-European, the pronoun-copula root identity is highly improbable under a closed-system model of root formation, supporting the hypothesis that the two paradigms share a single root rather than two independently originated morphemes that happen to be phonemically identical. In vertebrate genomics, the extreme conservation of ultraconserved elements is best understood as the product of strong constraints within a finite nucleotide alphabet rather than neutral sampling from an open mutational space. Additional examples — Hebrew morphology in NLP, language acquisition, and spontaneous communicative acts such as Louie the parrot’s alarm — reinforce the same pattern across domains.

The distinction between probability and statistics is not merely terminological. It determines which questions we can meaningfully ask and which answers are available. Frequentist approaches excel when repeated sampling is possible and the system is open or incompletely characterized. Classical probability excels when the universe of possibilities is finite, structured, and known in advance. Recognizing this boundary clarifies inferential practice across disciplines. In each case examined here, treating the problem as one of classical probability within a closed system clarified an inferential question that a purely statistical framing left underdetermined or unnecessarily complex: Hebrew NLP pipelines built on statistical disambiguation alone struggled for decades before joint morphological-syntactic inference resolved the core difficulty (Section 6.2); the PIE copula-pronoun identity has been debated for well over a century, predating the closed-system framing proposed here; and for ultraconserved elements, the two approaches converge on the same conclusion by different routes, with the closed-system framework offering a more transparent and model-independent grounding for a result statistical methods had already established (Section 5.1).

Future work can extend the framework in several directions: developing explicit probability models for other closed systems such as regulatory networks and cryptographic protocols; and exploring how children exploit probabilistic reasoning within closed symbolic domains during language acquisition. By taking closed symbolic systems seriously as a distinct category of inferential problem, we gain not only better explanations for specific historical puzzles but a more nuanced understanding of the relationship between evidence, structure, and inference itself. The universe, after all, is full of finite systems waiting to be understood on their own terms. For spontaneous, unrepeatable events, probability is the only way to eliminate the possibility of chance.

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